SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering
Program: MBA Teach (AI) and B Tech AI Semester: VI

Course: Agile Product Development with DevOps

Experiment No.01

PART A

Title: Simulation of a Mini Project Using Waterfall Model and Transition to Agile Methodology

Objective: Understand and compare traditional and Agile development frameworks.

Task:

1. Simulate a Mini Project: Use the Waterfall model for a small project (e.g., a library management system) and identify challenges.
   (Group of students (2/3) define problem statement and complete the task assigned)
2. Transition to Agile by defining iterative cycles and compare outcomes (time, collaboration, and adaptability).

Tools to Use

- **For Waterfall Simulation:**
  - Documentation tools (e.g., MS Word, Google Docs).
  - Diagram tools for system design (e.g., Lucidchart, draw.io).

- **For Agile Simulation: (Demonstrate the usage of tool in Exp no 2 only define and compare iterative lifecycle and compare outcomes)**
  - **Trello or Jira:** Manage product backlogs, sprints, and tasks.
  - **Collaboration Tools:** Slack or Microsoft Teams for stand-ups and reviews.

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PART B
(PART B : TO BE COMPLETED BY STUDENTS)
(Students must submit the soft copy as per following segments within two hours of the practical.)

Roll No. A139	Name: Aryan Turalkar
Class: B Tech AI & DS	Batch: 2
Date of Experiment:	Date of Submission:
Grade:	

Part 1: Waterfall Model Implementation

Title: Luxury Resort Brochure ('NIVASA')

Client: College Professor

Team Size: 3

Team Roles:

1. Developer: Aryan Turalkar
2. Business Analyst: Mayank Sawant
3. Project Manager: Sayali Rajput

1. Requirement Analysis
   The first phase focused on gathering and finalizing project requirements for the luxury resort brochure web application. The goal was to create a visually appealing and informative website that showcases resort facilities, accommodations, activities, and contact details.

2. System Design
   Based on finalized requirements, the overall system layout and structure were designed.
   Design Components:
   - Homepage with hero banner
   - Information Sections
   - Card Based layout for amenities and packages
   - Contact Section For user communication

3. Implementation
   The implementation phase involved developing the actual brochure website using front-end web technologies.
4. Testing
   After completing the development, testing was conducted to verify functionality and user experience.
5. Deployment & Maintenance
   Model Deployed after some minor ui ux changes and after approved by the client

Challenges Identified in Waterfall Model:

1. Changes after implementation were difficult to do
2. End time work made it too compact for features
3. Limited ui designs

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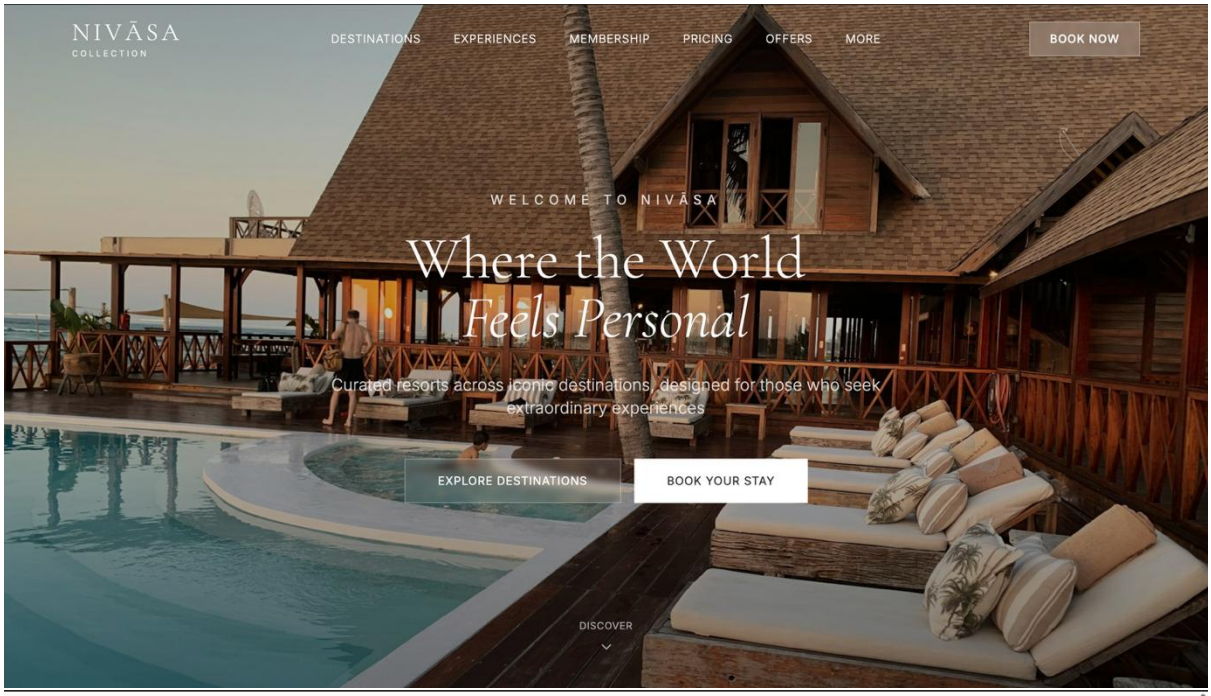

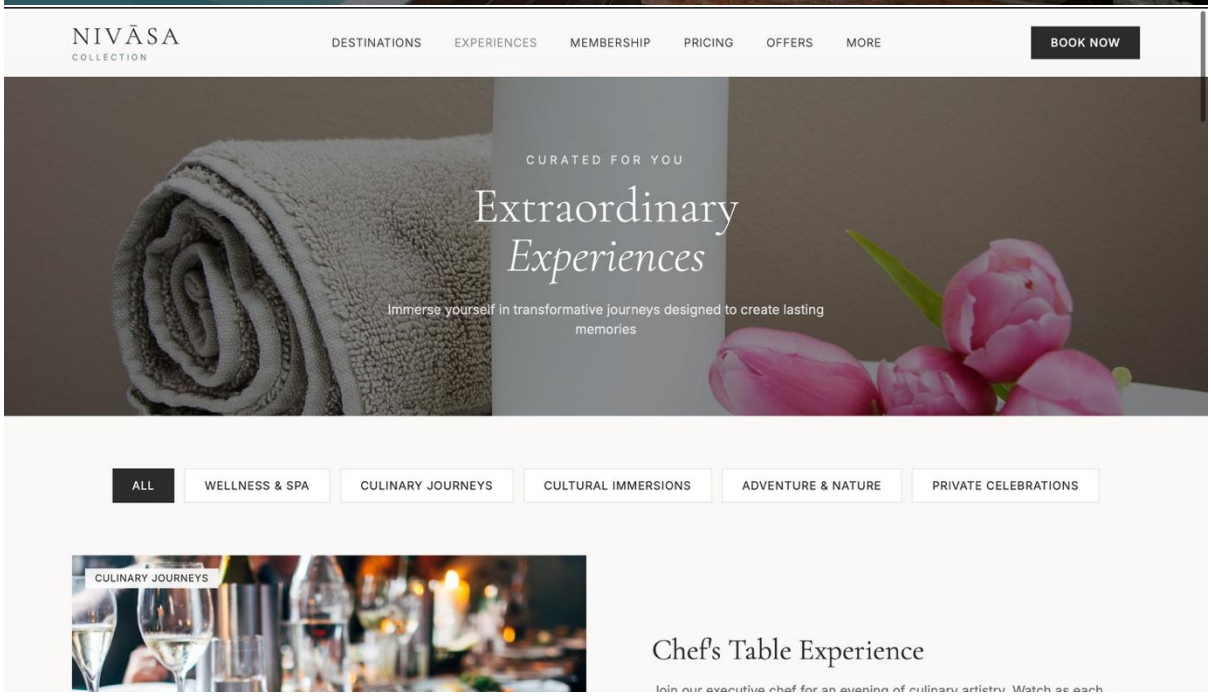

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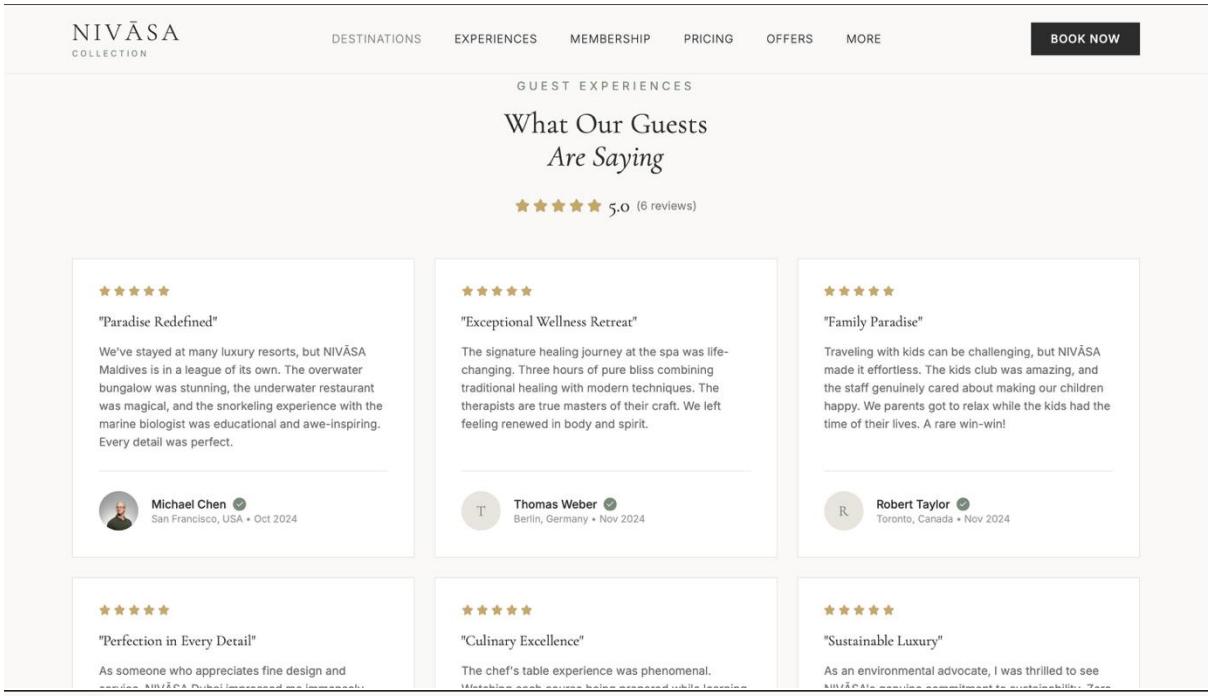

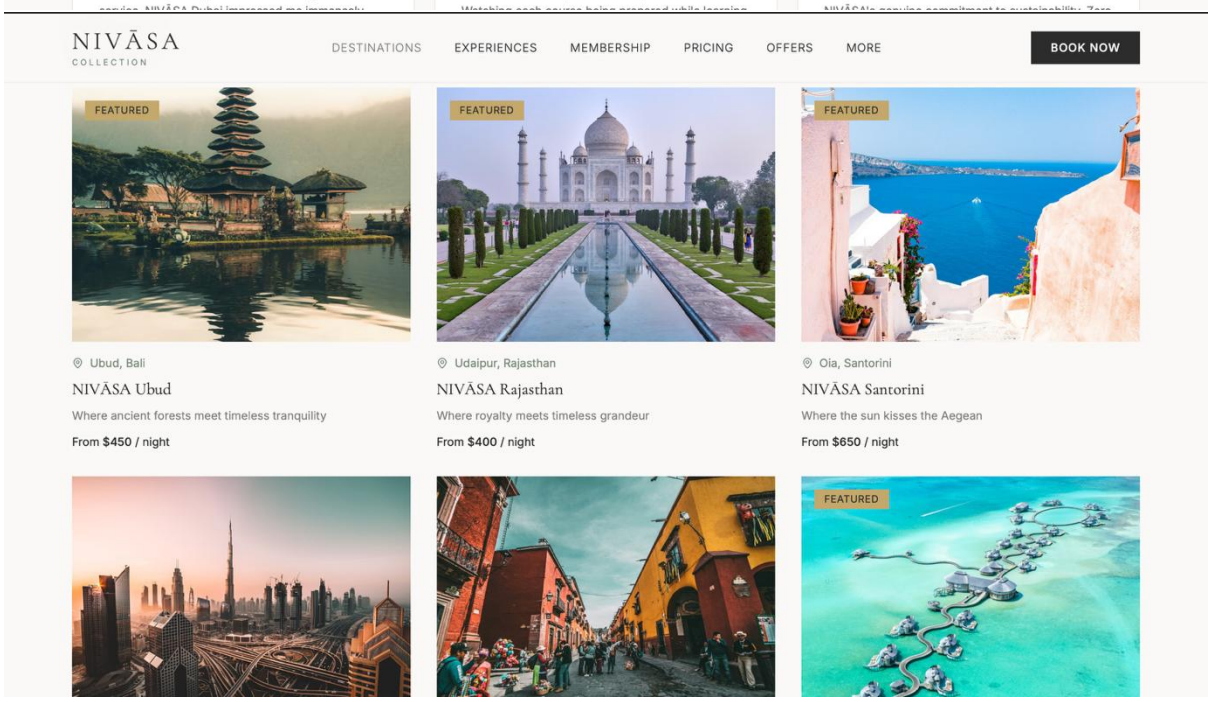

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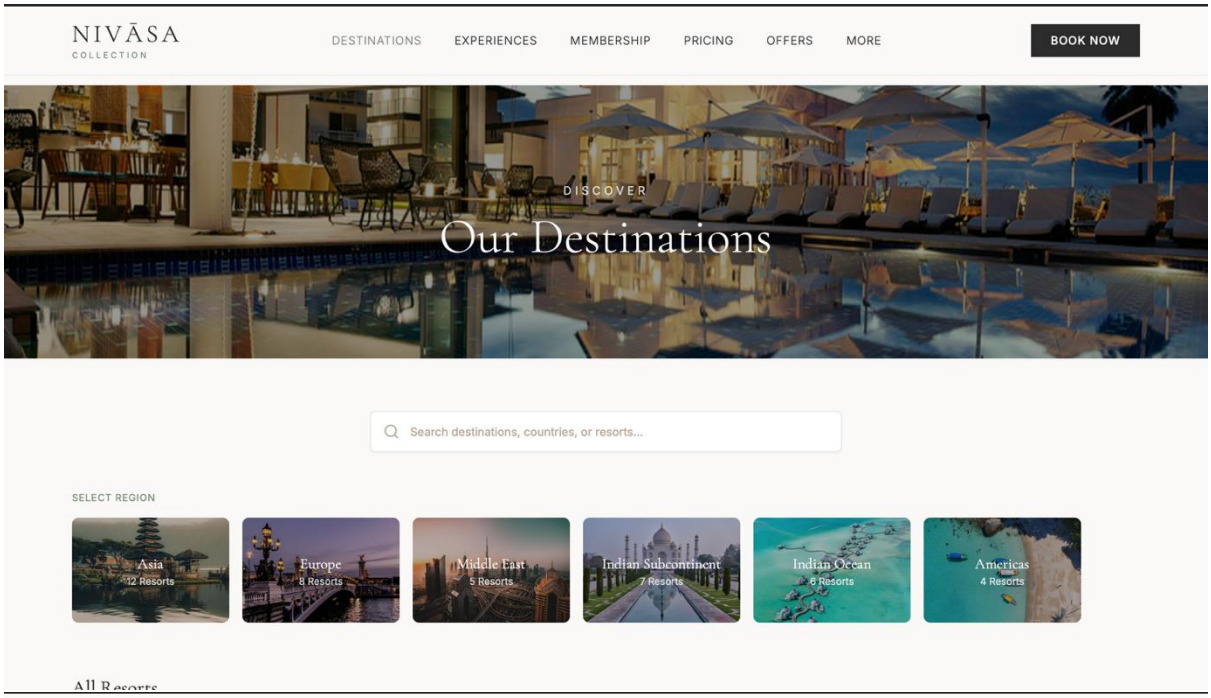

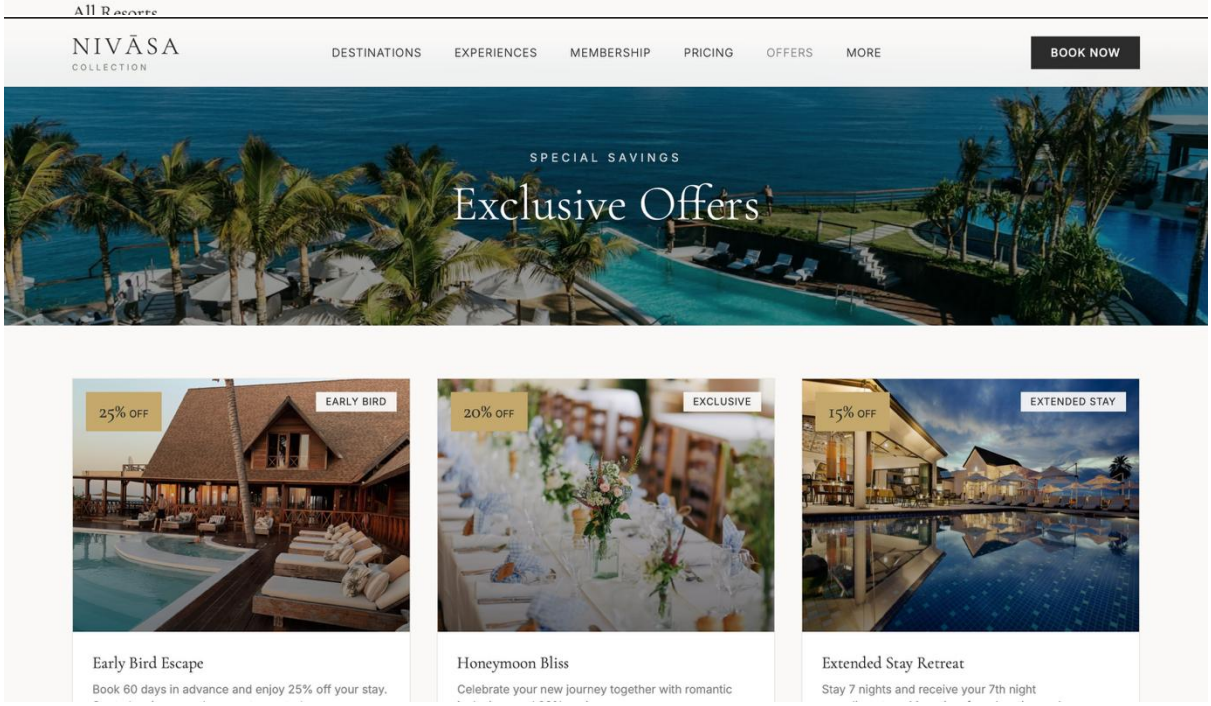

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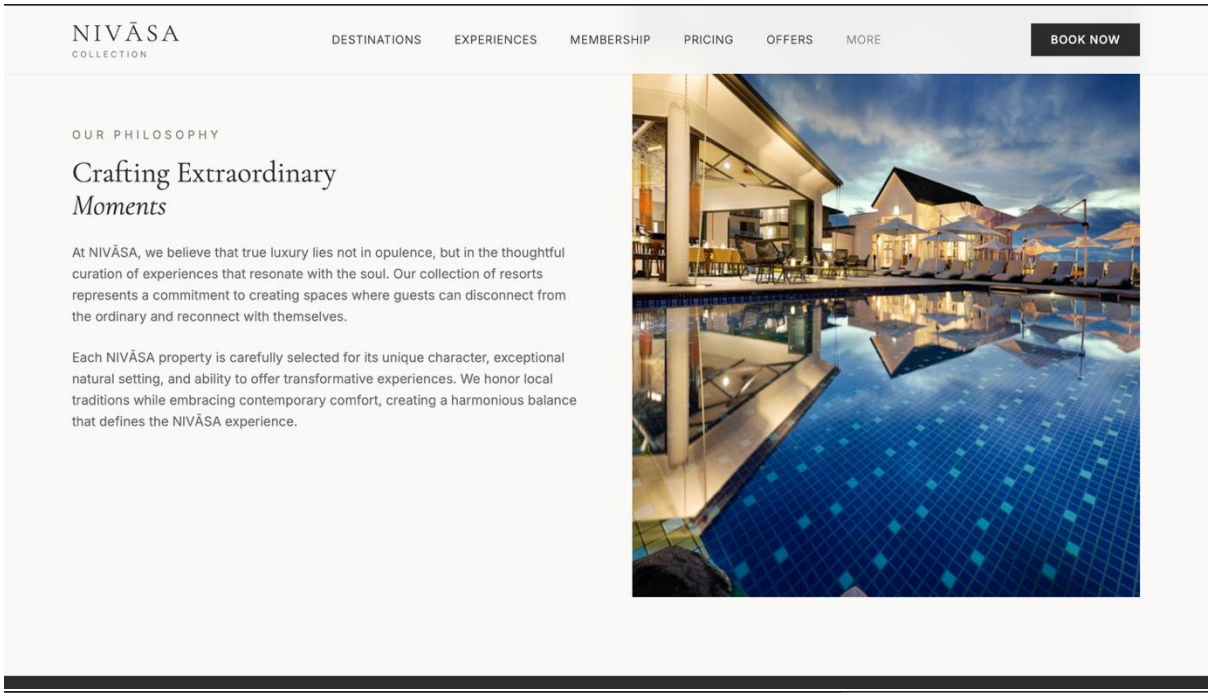

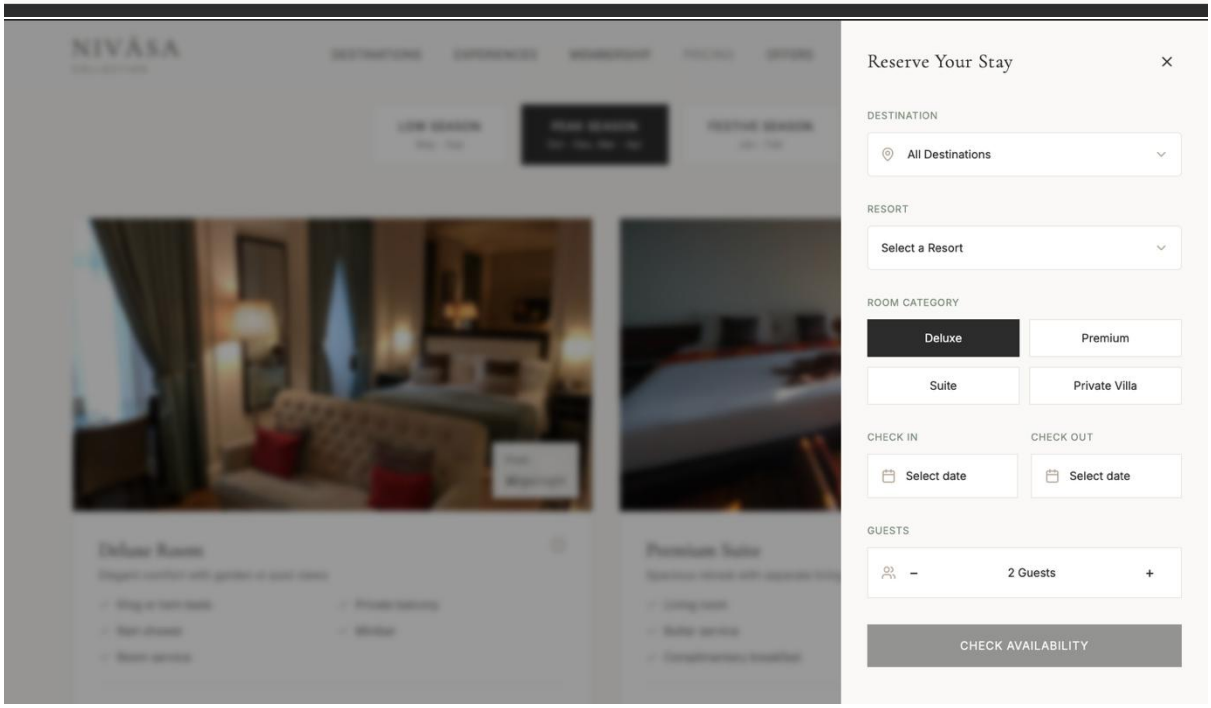

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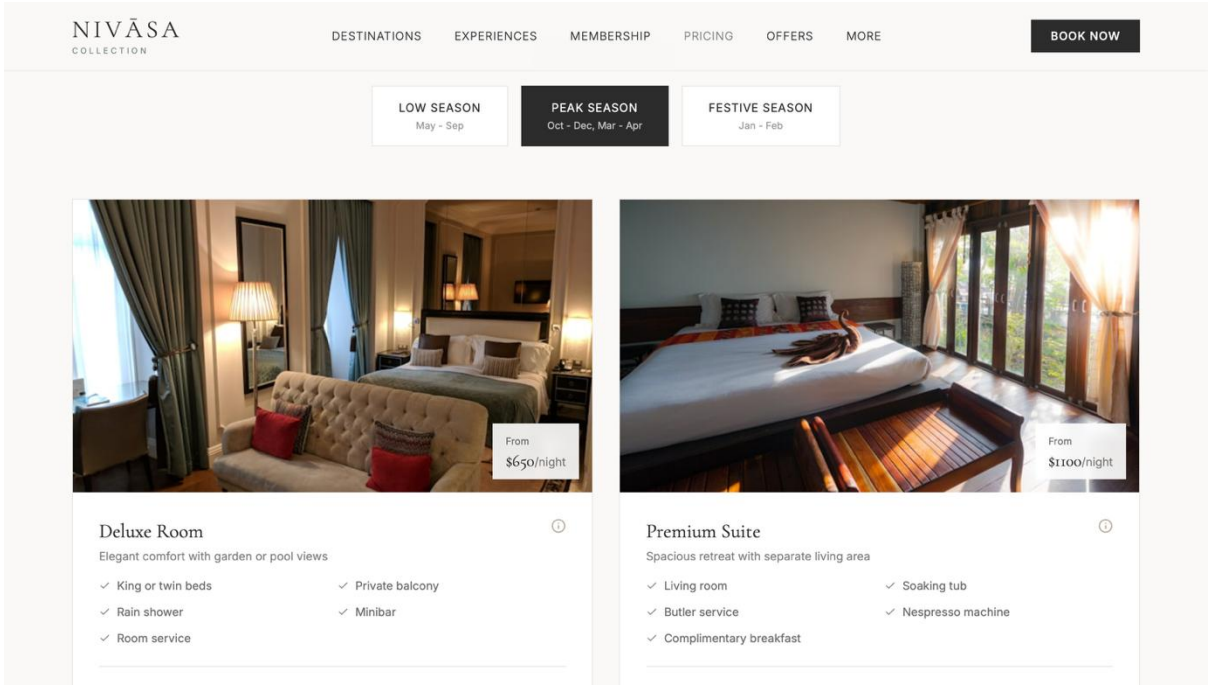

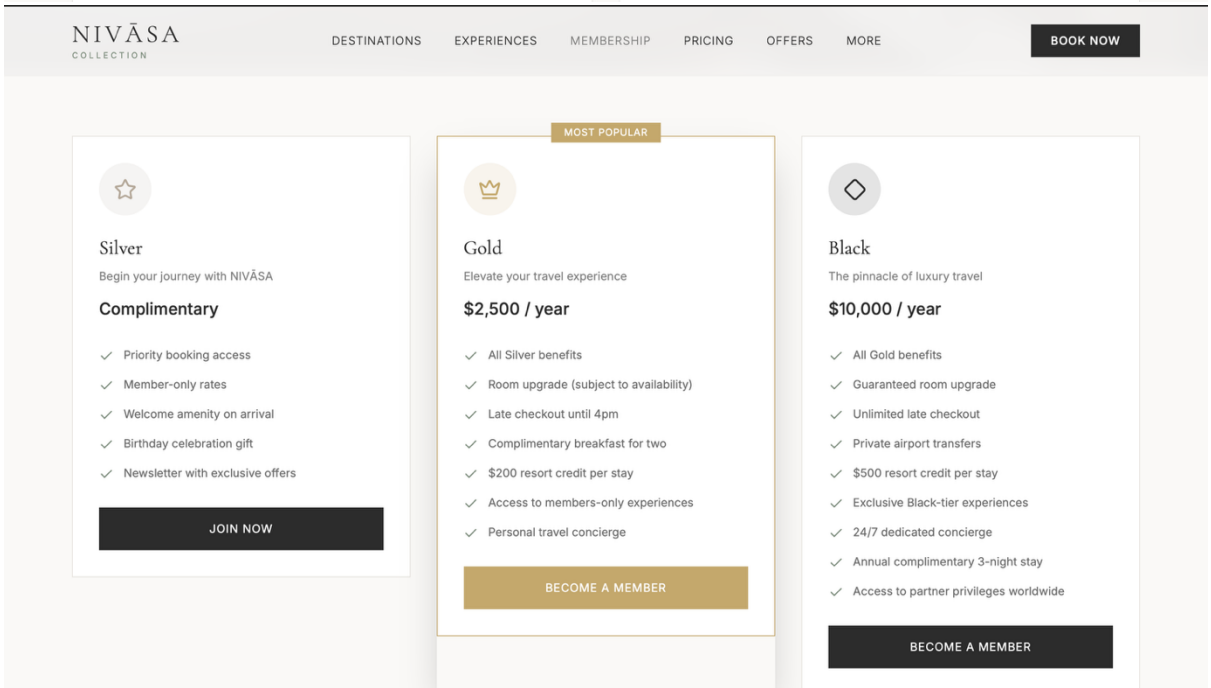

Part 2: Transition to Agile Methodology

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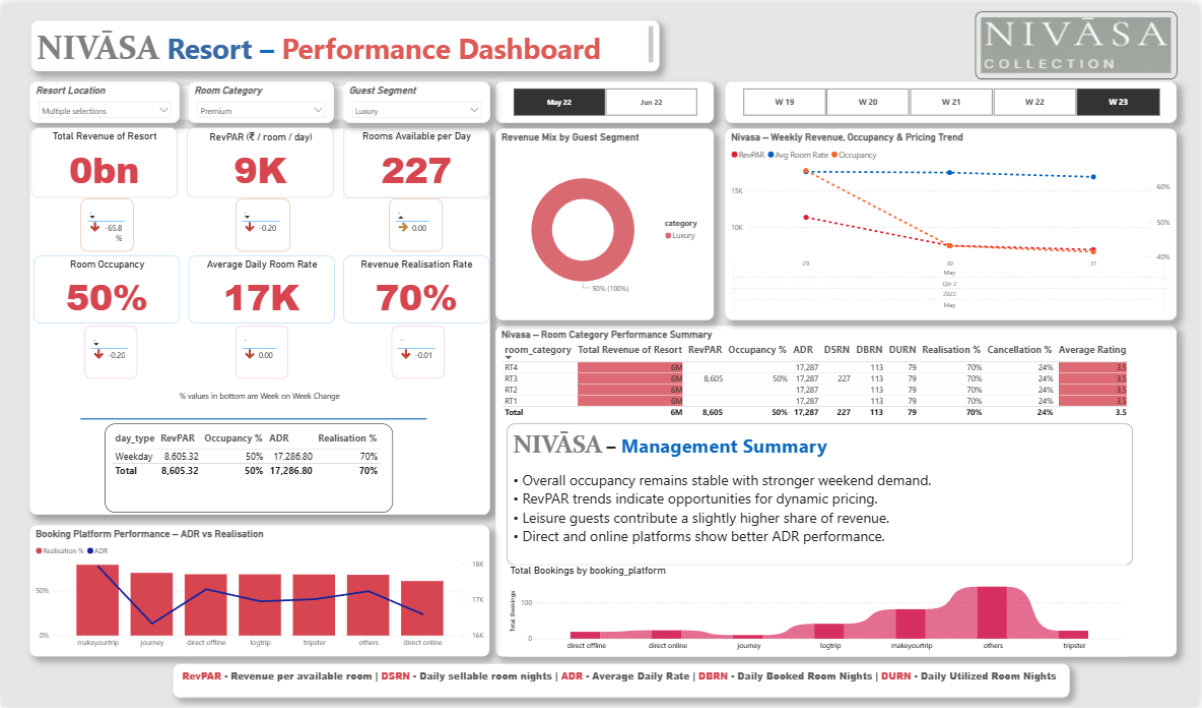

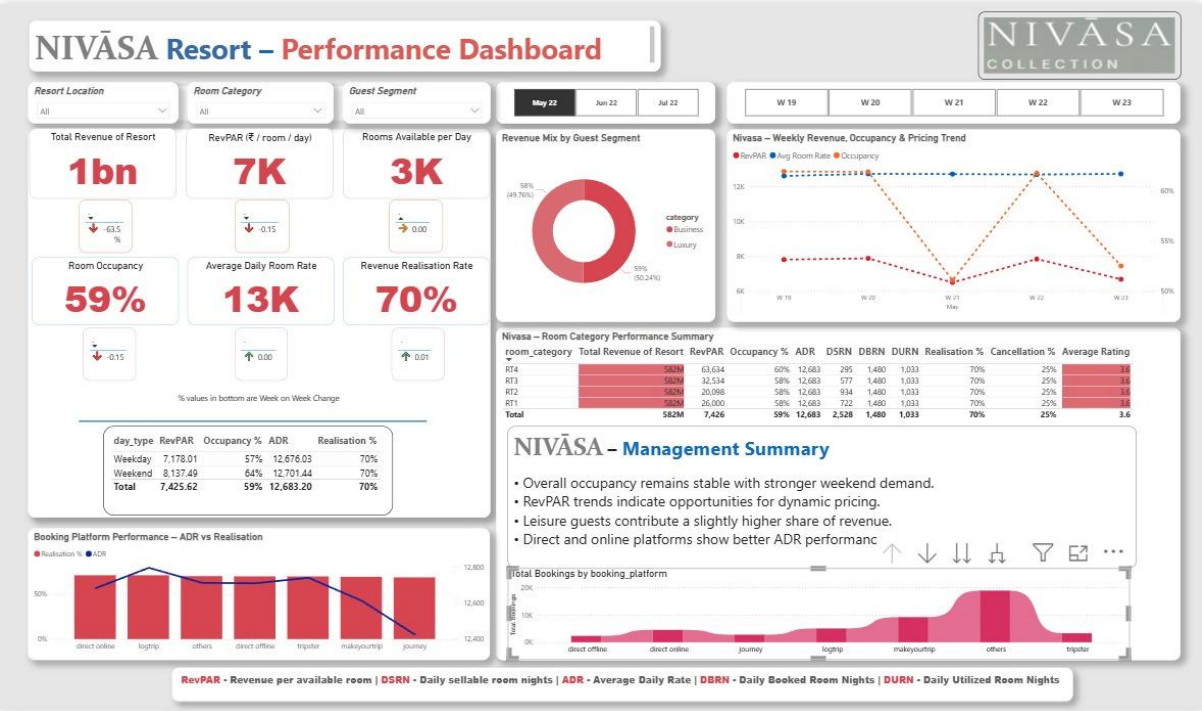

Conclusion

This experiment helped in understanding the application of the Waterfall model through the development of the NIVASA luxury resort brochure website. The structured approach supported clear requirement analysis and design, but limited flexibility when changes were required at later stages. Transitioning to the Agile approach highlighted how iterative planning and feedback can improve adaptability and collaboration. Overall, the activity provided practical insight into how different development models impact project execution and outcomes